Sahaj Bhadja

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Summary

Software engineer with 2+ years building AI-backed products, data pipelines, and backend integrations. Strong in Python, JavaScript/TypeScript, SQL, LLM workflows, testing, and reliable systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
M.S. in Computer Science *GPA: 3.75/4*
Coursework: Algorithms, Databases, Web Tech, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
B.Tech. in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Eng., OS, OOP, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected multi-tenant platform services and SQL-backed workflows for 500+ users, tuning indexed queries and worker flows across FastAPI services to cut dashboard latency from 1.2s to 350ms.
- Built fault-tolerant orchestration and validation flows for 1,000+ AI tasks and recurring integrations, improving throughput threefold while strengthening release reliability, debugging speed, and operational visibility.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered an AI-powered grading platform with Flask, React, OCR, and multimodal LLMs, supporting 10,000+ daily requests through secure uploads, grading workflows, and instructor-facing product features.
- Designed concurrent batch-processing for 1,000+ uploaded files using multithreading, AWS RDS, and CI/CD-backed releases, reducing feedback turnaround by 60% while maintaining stable APIs under peak demand.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Created Python evaluation pipelines across 100,000+ records, producing metrics, logs, and visual summaries that supported repeatable model analysis and quality review for AI and ML workflows.
- Documented prompt, evaluation, and reporting workflows for researchers, turning ad hoc experiments into reproducible processes with clearer traceability, faster handoffs, and more reliable analysis.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered API and ETL integrations with Python, SQL, AWS Glue, and S3, improving response time by 30% while supporting scalable service communication and maintainable backend components.
- Maintained 20+ automated tests and Jenkins pipelines with PyTest and Mockito, increasing CI reliability by 68% and strengthening release quality across deployments, debugging, and shared environments.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built full-stack product features with REST APIs, WebSockets, and responsive interfaces, supporting 2M+ synchronized records across modular workflows and reliable end-to-end application behavior.
- Automated Docker-based build and test flows with Jenkins CI/CD, cutting release time by 50% and improving deployment consistency, troubleshooting speed, and engineering collaboration across product environments.

Skills

Languages: Python, JavaScript, TypeScript, SQL, Java, C++, Bash
AI & Data: LLM workflows, prompt evaluation, OCR, semantic search, Pinecone, FAISS, PostgreSQL, MySQL, SQL Server, MongoDB
Backend: Flask, FastAPI, Node.js, Express, REST APIs, WebSockets, JSON, background jobs, ETL pipelines
Cloud & Delivery: AWS, Azure, GCP, AWS RDS, S3, Docker, Jenkins, Git, CI/CD, monitoring
Engineering: React, Next.js, unit testing, integration testing, debugging, code reviews, scalability, Agile/Scrum

Projects

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP*
- Built an AI-powered document platform for semantic search, summarization, and conversational QA over 50K+ records, combining vector retrieval, structured storage, and collaborative knowledge workflows.

Chat2SQL Code | *Java, TypeScript, SQL, Docker, Maven*
- Built an AI-powered SQL workspace supporting 10+ databases, accelerating natural-language query authoring, schema exploration, and structured data management through a unified Java and TypeScript app.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB, Plaid*
- Built a secure fintech platform supporting 50,000+ users and 1M+ transactions, combining real-time updates, scalable MongoDB access, and Kubernetes deployment for high-volume operational workflows.